ISyE 6669 FALL 2018 – SOLUTION SET FOR EXAM II

Question 1 (18 points). Consider the following LP problem:

$$
\begin{array}{llllcl}
\text{maximize} & -2x_1 & +3x_2 & -6x_3 & & \\
\text{subject to} & 3x_1 & +x_2 & +2x_3 & \geq & 1 \\
 & 4x_1 & -2x_2 & +x_3 & = & 5 \\
 & 3x_1 & +2x_2 & +x_3 & \leq & 9 \\
 & \multicolumn{5}{l}{x_1 \geq 0, \quad x_2 \geq 0, \quad x_3 \geq 0.}
\end{array} \tag{1}
$$

(a) (**8 points**) State the dual linear program.

(b) (**10 points**) Suppose someone tells you that, at optimality, the first and second dual variables are non-zero and the third dual constraint is not tight, ie., this constraint evaluated at the optimal solution is not satisfied as equality. Could this be correct? Explain.

Solution. (a) The dual is

$$
\begin{array}{llllcl}
\text{minimize} & y_1 & +5y_2 & 9y_3 & & \\
\text{subject to} & 3y_1 & +4y_2 & +3y_3 & \geq & -2 \\
 & y_1 & -2y_2 & +2y_3 & \geq & 3 \\
 & 2y_1 & +y_2 & +y_3 & \geq & -6 \\
 & \multicolumn{5}{l}{y_1 \leq 0, \quad y_2 \text{ urs}, \quad y_3 \geq 0.}
\end{array} \tag{2}
$$

(b) By the complementarity slackness condtion, if the first and second dual variables are non-zero, then an optimal solution (x_1, x_2, x_3) must satisfy

$$
\begin{aligned}
3x_1 + x_2 + 2x_3 &= 1 \\
4x_1 - 2x_2 + x_3 &= 5
\end{aligned}
$$

Also, if the third dual constraint is not tight, the n we must have $x_3 = 0$. Hence, we conclude that

$$
\begin{aligned}
3x_1 + x_2 &= 1 \\
4x_1 - 2x_2 &= 5
\end{aligned}
$$

and hence that $x_1 = 0.7$ and $x_2 = -1.1$. But since (x_1, x_2, x_3) must satisfy $x_2 \geq 0$, we obtain a contradiction. Hence, the claim in (b) is NOT possible. ■

Question 2 (32 points). During the next four months, a construction firm must complete three projects. Project 1 must be completed within three months and requires 8 months of labor. Project 2 must be completed within four months and requires 10 months of labor. Project 3 must be completed at the end of two months and requires 12 months of labor. Each month, 8 workers are available. During a given month, no more than 6 workers can work on a single job. Formulate a maximum flow problem that could be used to determine whether all three projects can be completed on time. DO NOT SOLVE THE PROBLEM. (*Hint:* If the maximum flow in the network is 30, all projects can be completed on time.)

Solution. ■

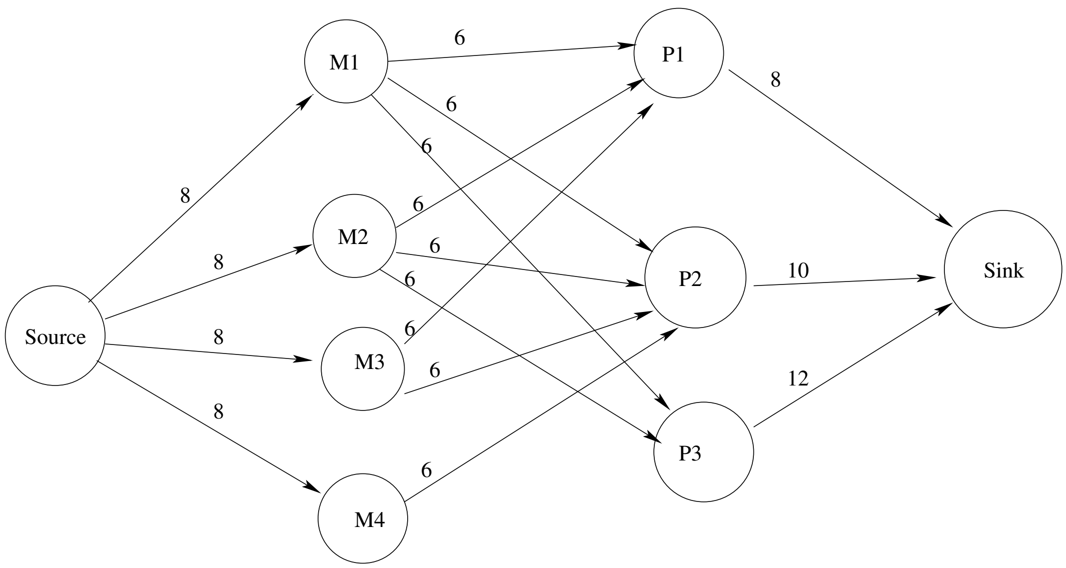

Figure 1: Maximum Flow Picture

Question 3 (50 points)**.** Consider the following LP:

$$
\begin{array}{lrrrrl}
\text{minimize} & z = & c_1x_1 & +c_2x_2 & +c_3x_3 & \\
\text{subject to} & & A_{11}x_1 & +A_{12}x_2 & +A_{13}x_3 & \geq\ b_1 \\
 & & A_{21}x_1 & +A_{22}x_2 & +A_{23}x_3 & \leq\ b_2 \\
 & & x_1 \geq 0, & x_2 \geq 0, & x_3 \geq 0. &
\end{array} \tag{3}
$$

and its associated optimal tableau

$$
\begin{array}{r|c|cccccc|c}
BV & z & x_1 & x_2 & x_3 & e_1 & s_2 & rhs \\
\hline
z & 1 & 0 & -3 & -3 & -2 & 0 & \bar{z} \\
\hline
x_1 & 0 & 1 & 2 & 1 & 1 & 0 & 8 \\
s_2 & 0 & 0 & 3 & -1 & 1 & 1 & 12
\end{array} \tag{4}
$$

The variables e_1 and s_2 in the above tableau are the excess variable for the first constraint of (3) and the slack variable for the second constraint of (3), respectively. Answer the following questions:

(a) (**6 points**) Compute the optimal basis A_B.

(b) (**11 points**) Compute the coefficients b_1, b_2, A_{11}, A_{12}, A_{13}, A_{21}, A_{22} and A_{23}.

(c) (**11 points**) Compute the coefficients c_1, c_2, c_3 and $\bar{z}$.

(d) (**7 points**) Find the dual of the linear program (3) and an optimal solution for this dual problem.

(e) (**7 points**) What is the range of values of b_1 for which the current set of basic variables remain optimal?

(f) (**8 points**) Solve the LP problem obtained by changing the current value of b_1 to 2.

Solution. Recall the fundamental formulas:

$$\bar{A} = A_B^{-1}A, \quad \bar{b} = A_B^{-1}b, \quad \bar{c}^T = c_B^T A_B^{-1}A - c^T, \quad \bar{z} = c_B^T A_B^{-1} b$$

(a) By the first fundamental formula, it follows that $A_B\left[\bar{A}_{e_1}, \bar{A}_{s_2}\right] = [A_{e_1}, A_{s_2}]$, i.e.

$$A_B \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

and hence that

$$A_B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix}.$$

(b) By the second fundamental formula, in particular $b = A_B\bar{b}$, we obtain

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 12 \end{bmatrix} = \begin{bmatrix} -8 \\ 4 \end{bmatrix}.$$

By the first fundamental formula, and the values in A_B in part (a), we obtain

$$\begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix} = A_B\left[\bar{A}_{x_1}, \bar{A}_{x_2}, \bar{A}_{x_3}\right] = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & -1 \end{bmatrix} = \begin{bmatrix} -1 & -2 & -1 \\ -1 & 1 & -2 \end{bmatrix}.$$

(c) By the fourth fundamental formula, we first observe that

$$[c_{x_2}, c_{x_3}] = [c_{x_1}, c_{s_2}]\left[\bar{A}_{x_2}, \bar{A}_{x_3}\right] - [\bar{c}_{x_2}, \bar{c}_{x_3}] = [c_{x_1}, 0]\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} - [-3, -3] = [2c_{x_1} + 3, c_{x_1} + 3].$$

Now since

$$0 = c_{e_1} = [c_{x_1}, c_{s_2}]\left[\bar{A}_{e_1}\right] - \bar{c}_{e_1} = [c_{x_1}, 0]\begin{bmatrix} 1 \\ 1 \end{bmatrix} - (-2) = c_{x_1} + 2,$$

which implies $c_{x_1} = -2$, we obtain

$$[c_1, c_2, c_3] = [-2, 2(-2) + 3, -2 + 3] = [-2, -1, 1].$$

As a consequence, the optimal solution value can be computed as

$$\bar{z} = c_B^T\bar{b} = [-2, 0]\begin{bmatrix} 8 \\ 12 \end{bmatrix} = -16.$$

(d) The dual of the linear program (2) is

$$\begin{array}{ll} \text{maximize } z = & -8y_1 + 4y_2 \\ \text{subject to} & -y_1 - y_2 \le -2 \\ & -2y_1 + y_2 \le -1 \\ & -y_1 - 2y_2 \le 1 \\ & y_1 \ge 0, y_2 \le 0. \end{array}$$

An optimal solution for this dual problem is

$$\bar{\pi}^T = (y_1^*, y_2^*) = c_B^T A_B^{-1} = \begin{bmatrix} -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \end{bmatrix}.$$

(e) The current set of basic variables remain feasible as long as

$$0 \le \bar{b} = A_B^{-1}b = \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ 4 \end{bmatrix} = \begin{bmatrix} -b_1 \\ -b_1 + 4 \end{bmatrix}$$

and hence we require $b_1 \in (-\infty, 0]$.

(f) With $b_1 = 2$ the new values of $\bar{z}$ and $\bar{b}$ become

$$\bar{b}_{\text{new}} = (-2, 2)^T, \qquad \bar{z}_{\text{new}} = c_B^T \bar{b}_{\text{new}} = (-2, 0)(-2, 2)^T = 4,$$

and the new canonical tableau becomes:

BV	z	x_1	x_2	x_3	e_1	s_2	rhs
z	1	0	-3	-3	-2	0	4
x_1	0	1	2	1	1	0	-2
s_2	0	0	3	-1	1	1	2
ratio	X	X	X	X	X	X	

From the above, x_1 must leave the basis. However, since there are no values in row 1 which are negative, we must conclude that this new LP is infeasible.